

Chapter 1

Week 11 — File Handling, Serial Ports, Graphics & Debugging

ENEL712 Embedded Systems Design — Lecture Week 11

Summary

Week 11 wider summary: file handling and serial ports, GUI graphics and plotting, Visual Studio debugging tools, and a refresher on core programming concepts.

Topics Covered

- File handling and serial ports
- Graphics and plotting
- Debugging in Visual Studio
- OOP / C# review concepts

1.1 File Handling and Serial Ports

- Files are sequential byte streams ending in EOF.
- Text files (ASCII/Unicode) vs binary files (raw bytes).
- StreamReader/StreamWriter handle text + encoding.
- FileStream is the byte-level base class; requires a FileMode (Append/Create/Open).
- BinaryReader/BinaryWriter read/write structured binary formats (JPEG, WAV, PNG).
- OpenFileDialog and SaveFileDialog: standard Windows file pickers; configurable Filter property.

1.1.1 SerialPort communication

- Object simplifies COM port use; create programmatically or in the designer.
- Methods Open / Close / Read / Write are blocking by default.

- Watch buffering, keep synchronisation with remote device, handle endianness (Intel = Little-Endian, AVR = Big-Endian).

1.2 Graphics and Plotting

- PictureBox displays GIF/JPG/BMP/PNG images.
- Custom drawing requires a surface (Panel/Form), a Graphics object, and a Pen.
- Paint event keeps custom drawings persistent — they vanish on minimise/cover unless redrawn.
- Pen draws lines and shape borders (Width, DashStyle, Transparency).
- Brush fills shapes (SolidBrush, TextureBrush).
- Graphics methods: DrawRectangle, FillEllipse, DrawString, etc.
- Chart control auto-creates plots (default bar) and supports line charts via the Series Collection Editor.

1.3 Debugging in Visual Studio

- Breakpoints suspend execution at a chosen line for inspection.
- Inspection windows: Autos/Locals show current-scope variables; Watch lets you track expressions.
- Call Stack shows the function chain that led to the current breakpoint.
- Stepping: F10 (Step Over), F11 (Step Into), Shift+F11 (Step Out).
- Conditional breakpoints trigger only when a C# expression is true.
- System.Diagnostics.Debug.WriteLine() prints to the Output window without breakpoints.

1.4 OOP / C# Review Concepts

- Use the smallest data type sufficient for your data.
- Control flow: for, while, if/else.
- Functions: return type, name, args; pointers/handles allow extra outputs.
- Pointers (*x) and handles (&x) let functions modify the caller's variables.
- Arrays are 0-indexed sequences of one type.
- Dynamic memory allocation: new/delete in C/C++ (manual) and new + GC in C# (automatic).

1.5 Use Cases & Applications

- Lab tooling: a single WinForms app combining a SerialPort + Chart + log file is a standard 'desktop companion' for an embedded device.
- Async IO: when listening to multiple ports or sockets, await over Read/Write keeps the UI responsive without manual threading.

- Visual Studio's debugger is the most efficient way to learn unfamiliar libraries — set a breakpoint and inspect.
- PictureBox + Paint event is enough to render real-time waveforms without bringing in a charting library.

1.6 Worked Example: Logging Serial Data to a File

```
private SerialPort sp;
private StreamWriter logger;

void StartLogging(string port, string path)
{
    logger = new StreamWriter(path, append: true);
    sp = new SerialPort(port, 115200);
    sp.DataReceived += (s, e) =>
    {
        string chunk = sp.ReadExisting();
        logger.Write(chunk);
        logger.Flush();
        Invoke((Action)(() => logTBox.AppendText(chunk)));
    };
    sp.Open();
}
```

1.7 Plotting Live Data on a PictureBox

```
private readonly Queue<float> samples = new();

private void picBox_Paint(object sender, PaintEventArgs e)
{
    var g = e.Graphics;
    var pen = new Pen(Color.SteelBlue, 1.0f);
    int x = 0;
    float[] arr = samples.ToArray();
    for (int i = 1; i < arr.Length; i++, x++)
        g.DrawLine(pen,
                    x, picBox.Height - arr[i - 1],
                    x + 1, picBox.Height - arr[i]);
}

void AddSample(float v)
{
    samples.Enqueue(v);
    while (samples.Count > picBox.Width) samples.Dequeue();
    picBox.Invalidate(); // schedule a Paint redraw
}
```

1.8 Debugging Cheat Sheet

Action	Shortcut	Use
Toggle breakpoint	F9	Pause execution at a line
Step Over	F10	Run current line; don't enter called methods
Step Into	F11	Enter called method
Step Out	Shift+F11	Run to end of current method, return to caller
Continue	F5	Run until the next breakpoint or exception
Conditional break	F9 → right-click → Conditions	Pause only when an expression is true

Key Definitions

Stream

Sequential byte flow used for file/IO.

EOF

End-of-file marker or condition.

Text File

Human-readable file (ASCII/Unicode).

Binary File

Raw bytes interpreted with a known data format.

FileMode

Enum used by FileStream (Open/Create/Append).

Blocking Method

Function call that halts execution until done.

Little-Endian

LSB stored at the lowest address (Intel).

Big-Endian

MSB stored at the lowest address (AVR / network).

Paint Event

GUI event firing when a window must be redrawn.

Graphics Object

Provides drawing methods like DrawRectangle, FillEllipse, DrawString.

Pen Object

Tool for drawing lines and outlines (width, style, colour).

Brush Object

Tool defining the fill of shapes (SolidBrush, TextureBrush).

Breakpoint

Marker that suspends program execution for inspection.

Call Stack

Window showing function call history at a breakpoint.

Pointer (*x)

Variable holding an address; enables pass-by-reference.

Dynamic Memory Allocation

Allocating memory at runtime, not compile time.

Invalidate

Tells a control to redraw itself the next time the message loop processes paint events.

Conditional Breakpoint

Breakpoint that pauses only when a Boolean expression evaluates to true.

Output Window

Visual Studio panel that receives Debug.WriteLine messages while the program runs.

Trace.Listeners

Mechanism for routing trace messages to files, consoles, or other sinks for production diagnostics.

Sources

- Lecture Week 11.pdf